



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. IT

Course Code & Title: CS3352 & Foundations of Data Science

Name of the Faculty member (s): Dr. V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Describing Data with Averages
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** Jigsaw
- **Justification:**

Describing data using averages is a key concept in data science. Different problem sets were assigned to various groups. The primary goal of this Jigsaw activity is to enhance individual and peer learning by encouraging students to thoroughly understand the concept and engage in collaborative learning.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The topic "Describing Data with Averages" was taught in class.
 - The class was divided into 21 teams, each consisting of 3 members, grouped based on academic performance and ability. These groups were called home groups.
 - One student from each home group was selected as a leader, and expert groups were formed, comprising one student from each home group.
 - Each student in a group was assigned a specific topic, with 10 minutes allocated for discussion. The course instructor provided instructions on how to carry out this activity.
 - For collaborative teamwork, the expert groups were dissolved, and all students returned to their respective home groups.
 - In the home groups, each member became an expert in one of the hashing resolution techniques. They shared their ideas with the rest of the group, enabling every student to learn all the concepts.
 - This approach helped students gain in-depth knowledge of the topic. Afterward, they discussed and answered the given questions. as shown in below the and the discussion as shown in figure 1.

Question:

1. a) Calculate the standard deviation and mean of a dataset with values:
[5, 8, 12, 15, 20].
b) Find the mode of the following numbers: 5, 8, 6, 9, 8, 3, 5, 2, 8.

2. a) Calculate the median for the given grouped data:
[10-20]: 5, [20-30]: 12, [30-40]: 8, [40-50]: 6.
b) Calculate the mode of the following test scores: 88, 92, 78, 92, 85, 90, 88, 93.
3. a). In a survey, participants' ages are recorded. Find the median age from the following ages: 29, 31, 35, 40, 41, 45, 49.
b) . Find the standard deviation for a dataset with values: [10, 10, 10, 10, 10].
4. a) . Find the median for the following dataset: [0-10]: 15, [10-20]: 25, [20-30]: 10.
b) Calculate the mode of the following test scores: 88, 92, 78, 92, 85, 90, 88, 93.
5. a) . Determine the standard deviation for a dataset with values: [2, 4, 6, 8, 10]. b) In a survey, participants' ages are recorded. Find the median age, mean from the following ages: 29, 31, 35, 40, 41, 45, 49.
6. a) What is the mode of the shoe sizes for a group of 20 people: 8, 9, 10, 9, 7, 8, 9, 10?
b) Determine the median for grouped data: [0-5]: 8, [5-10]: 12, [10-15]: 20, [15-20]: 15.
7. a) Calculate the standard deviation of a sample with values: [12, 14, 18, 20, 22].
b) Calculate the mean, median for the following dataset of annual incomes: \$45,000, \$50,000, \$55,000, \$60,000, \$75,000, \$80,000, \$100,000.
8. a) Find the median for grouped data: [0-100]: 7, [100-200]: 18, [200-300]: 11.
b) Calculate the mode of the temperatures recorded for a week: 78°F, 82°F, 78°F, 77°F, 83°F, 79°F, 78°F.
9. a) . Determine the median for the grouped data: [0-25]: 3, [25-50]: 9, [50-75]: 16.
b) . Find the mode of the following series: 12, 18, 24, 12, 30, 18, 36, 24.
10. a) Find the standard deviation for a dataset with values: [3, 7, 11, 15, 19]. b) . Find the median height of a group of individuals with the heights (in inches): 62, 64, 66, 68, 70, 72, 74.
11. a) A data set represents the number of hours worked by employees in a week: 32, 40, 42, 45, 50, 55. Calculate the median.
b) Calculate the population standard deviation for a dataset with values: [8, 16, 24, 32, 40].
12. a) Determine the median for the grouped data: [0-25]: 3, [25-50]: 9, [50-75]: 16.
b) . Determine the mode of the following list of book prices: \$12.50, \$15.75, \$12.50, \$18.25, \$20.50, \$15.75.
13. a) Find the median height of a group of individuals with the heights (in inches): 62, 64, 66, 68, 70, 72, 74.
b) Determine the standard deviation for a dataset with values: [5, 12, 19, 26, 33].
14. a) A data set represents the number of hours worked by employees in a week: 32, 40, 42, 45, 50, 55. Calculate the median.
b) Calculate the standard deviation of a dataset with values: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10].
15. a) . Calculate the median for the dataset grouped as [0-5]: 12, [5-10]: 14, [10-15]: 8, [15-20]: 11.
b) . Calculate the mode of the number of siblings for a group of 50 students: 1, 2, 2, 0, 1, 3, 1, 0.
16. a) Find the standard deviation for a dataset with values: [0, 5, 10, 15, 20, 25, 30, 35, 40, 45].
b) From a set of daily website visit counts: 200, 250, 300, 350, 400, 450, 600, find the median.
17. a) . Find the median for the grouped data: [0-30]: 20, [30-60]: 15, [60-90]: 10.
b) Find the mode of the ages of a group of people: 35, 27, 41, 35, 29, 29, 41, 35, 38.
18. a) . Determine the median for the dataset grouped as [0-40]: 6, [40-80]: 12, [80-120]: 18.
b) Determine the mode of the number of daily website visitors over a week: 100, 150, 125, 150, 100, 175, 200.

19. a) Calculate the standard deviation of a sample with values: [22, 24, 25, 28, 30, 32].
b) Calculate the mean, median for a dataset representing the response times (in seconds) for a web application: 3.2, 4.5, 5.1, 5.8, 6.4, 7.0.
20. a) . Calculate the mode of the monthly rainfall measurements (in inches) for a year: 2.5, 3.0, 1.5, 2.5, 1.0, 3.5, 2.5, 1.5, 3.0.
b) . Determine the standard deviation for a dataset with values: [2, 2, 2, 8, 8, 8].
21. a) Calculate the median for the following grouped data:
[0-10]: 7, [10-20]: 14, [20-30]: 5.
b) . Find the mode of the following numbers: 4, 5, 6, 6, 8, 8, 8, 10, 11, 12.

• **Images / Screenshot of the practice:**



Figure 1: Students discussion with their team members

• **CO – PO / PSO mapping: PO / PSO mapped:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	2	1	2	2	3	2

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO1	PSO2
	3	3	2	2	2	1	2	2	3
Justification for correlation	Apply basic Knowledge and fundamentals of Mathematics for solving practical problems	Identify the need of describing data for developing applications	Able to design solutions for applications to describe the data and find the relation between data	Able to analyze the relationship between data and provide valid conclusions based on Describing data	Ability to describe data individually or teamwork	Ability to recognize the need for describing averaging techniques in real world applications	Exhibit knowledge in describing averaging techniques to provide solution for complex computational problems	Exhibit knowledge in describing averaging techniques in the field of IoT, Cloud Computing and Data Analytics to provide IT solutions.	Exhibit knowledge in describing averaging techniques in software projects to deliver a quality product

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- All the students actively participated, completed the task, and practiced with their group members.
- Each student shared and discussed their knowledge of mean, median and mode with others in the class.
- This activity helped them enhance their communication skills, including effectively interacting with their group and summarizing the concepts.

- ❖ ***Benefit of the practice:***

- The students actively participated in this activity.
- Through this activity, they learned how to communicate with others, collaborate as a team, and share their ideas. This emphasized the lesson that teamwork leads to better solutions.
- Even slow learners were able to grasp the concepts of mean, median, and mode effectively.

- ❖ ***Challenges faced in implementation:***

- Organizing the students into groups for this activity was time-consuming and challenging.
- Some team members did not actively participate in the group discussions.

References:

- ❖ <https://www.jigsaw.org/>

- ❖ <https://www.readingrockets.org/classroom/classroom-strategies/jigsaw>

Signature of Faculty Member

(Dr.V. Anusuya)

HOD